Type MS · Finding a point from a line and a distance

Answer 1 of 10 · Question 6 part d · 7 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part d in this mark scheme extract

Question	Scheme	Marks
6 (a)	$p=1,\ \ q=8$	B1B1 [2]
(b)	$\left(\text{Gradient of } l=\dfrac{12-2}{3--2}=\right)2$	B1
	(Gradient of the perpendicular =) $-\dfrac{1}{"2"}$	B1ft
	Equation of k	
	$y-"8"="-\dfrac{1}{2}"(x-"1")$	M1
	$\Rightarrow 2y+x-17=0$ *	A1 cso [4]
(c)	When $y=0$, $(2\times 0+x-17=0\Rightarrow)\ x=17$	B1
	$(\lvert CD\rvert=)\sqrt{(0-"8")^2+("17"-"1")^2}=\sqrt{320}=\left[8\sqrt{5}\right]$	M1A1 [3]
(d)	(Length of CX =) $\dfrac{2\times 80}{8\sqrt{5}}\left(=4\sqrt{5}\right)$	B1
	$\left("4\sqrt{5}"\right)^2=(m-"1")^2+(n-"8")^2$	M1
	$"2"=\dfrac{n-"8"}{m-"1"}\Rightarrow[n=2m+6]$ oe **or** $y-2="2"(x+2)\Rightarrow[y=2x+6]$	M1
	or $y-12="2"(x-3)\Rightarrow[y=2x+6]$ oe	
	$\Rightarrow "80"=(m-"1")^2+(2m+6-"8")^2\ (\Rightarrow 0=5m^2-10m-75$ oe eg $0=m^2-2m-15)$	ddM1
	$\Rightarrow(5m+15)(m-5)=0$ oe eg $(m+3)(m-5)=0$	M1
	$m=5,(-3)$	A1
	$n(=2m+6=2\times 5+6)=16$	A1 [7]
ALT1	(Length of CX =) $\dfrac{2\times 80}{8\sqrt{5}}\left(=4\sqrt{5}\right)$	B1
	$(m-"17")^2+(n)^2=\left("4\sqrt{5}"\right)^2+\left("8\sqrt{5}"\right)^2$	M1
	$"2"=\dfrac{n-"8"}{m-"1"}\Rightarrow[n=2m+6]$ oe **or** $y-2="2"(x+2)\Rightarrow[y=2x+6]$	M1
	or $y-12="2"(x-3)\Rightarrow[y=2x+6]$ oe	
	$\Rightarrow "400"=(m-"17")^2+(2m+6)^2\ (\Rightarrow 0=5m^2-10m-75$ oe eg $0=m^2-2m-15)$	ddM1
	$\Rightarrow(5m+15)(m-5)=0$ oe eg $(m+3)(m-5)=0$	M1
	$m=5,(-3)$	A1
	$n(=2m+6=2\times 5+6)=16$	A1 [7]

Type MS · Finding a point from a line and a distance

Answer 1 of 10 · Question 6 part d · 7 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part d in this mark scheme extract

ALT2	(Length of CX =) $\frac{2\times 80}{8\sqrt{5}}\left(=4\sqrt{5}\right)$	B1
	Length of AB = $\sqrt{(12-2)^2+(3--2)^2}\left(=5\sqrt{5}\right)$	M1
	Vector $\overrightarrow{AB}=\begin{bmatrix}3--2\\12-2\end{bmatrix}\left(=5\begin{bmatrix}1\\2\end{bmatrix}\right)$	M1
	$\left(\left\|\overrightarrow{CX}\right\|=4\sqrt{5}\Rightarrow\right)\overrightarrow{CX}=4\begin{bmatrix}1\\2\end{bmatrix}=\begin{bmatrix}4\\8\end{bmatrix}$	ddM1
	$\Rightarrow$ Coordinates of X are $(1,8)+(4,8)=(5,16)$	M1A1
	$(m=5 \quad n=16)$	A1 [7]
ALT3	$\frac{1}{2}\begin{vmatrix}17 & m & 1 & 17\\0 & n & 8 & 0\end{vmatrix}$	B1
	$\frac{1}{2}\begin{vmatrix}"17" & m & "1" & "17"\\0 & n & "8" & 0\end{vmatrix}=80\Rightarrow\frac{1}{2}\left(17n+"8"m-("136"+n)\right)=80\left(\Rightarrow 2n+m=37\right)$	M1
	$"2"=\frac{n-8}{m-1}\Rightarrow[n=2m+6]$ oe **or** $y-2="2"(x+2)\Rightarrow[y=2x+6]$ oe **or** $y-12="2"(x-3)\Rightarrow[y=2x+6]$ oe	M1
	$\frac{1}{2}\begin{vmatrix}"17" & m & "1" & "17"\\0 & "2m+6" & "8" & 0\end{vmatrix}$ or $\frac{1}{2}\left(17("2m+6")+"8"m-("136"+"2m+6")\right)=80$	ddM1
	or $2("2m+6")+m=37\Rightarrow m=$	M1
	$m=5,\ n=16$	A1A1 [7]
		Total 16 marks

Type MS · Finding a point from a line and a distance

Answer 1 of 10 · Question 6 part d · 7 marks

2024 JUN · Paper 01 · extract 3 of 5 · find part d in this mark scheme extract

Part	Mark	Notes
(a)	B1	For $p = 1$ OR $q = 8$ Allow anything that clearly implies these values eg (1, 8) or 1, 8
	B1	For $p = 1$ AND $q = 8$ Allow anything that clearly implies these values eg (1, 8) or 1, 8
(b)	B1	For the gradient of $l = 2$
	B1ft	For the gradient of $k = -\frac{1}{"2"}$. Ft their calculation of the gradient of l.
	M1	For the correct unsimplified equation for k, using their p and q and any **changed** gradient. If the candidate uses $y = mx + c$, a fully correct rearrangement to find c must be shown (this can be implied by a correct c for their equation) and concluded with the equation of the line written.
	A1 cso	For the equation of k in the required form. Minimum steps shown, no errors or omissions. This must from a correct p and q
Watch for methods using vectors. If the candidate arrives at the correct unsimplified equation with no obviously incorrect work, following vector work, allow B1 B1ft M1 and the final A1 if the equation is given in the correct form. (See example in Practice)		
(c)	B1	For the x coordinate of $D = 17$
	M1	For a correct **method** to find the length of CD using their p and q and their 17
	A1	For the correct length of $\sqrt{320}$ oe
(d)	B1	Any correct unsimplified calculation for CX
	M1	For correctly using Pythagoras to form an equation in terms of m and n or x and y. Follow through their CX (ie coming from correct calculation with their p, q, their point D).
	M1	For an unsimplified equation in terms of m and n or x and y correctly using the gradient of l. Follow through their gradient for l, their p and their q. Allow working in p, q or x, y Note this work is sometimes being seen in (b). **As long as it is used in part (d),** the equation found in (b) may be given credit.
	ddM1	For eliminating either m or n from either equation and forming a 3TQ. Allow errors in processing if an initial correct method following from 2 correct unsimplified equations to eliminate one of the variables is seen. Allow working in p, q or x, y Dependent on both previous method marks. Note, the previous M mark is for the unsimplified equation, so if simplified incorrectly, this can be used here to gain this mark (general principle of marking).
	M1	For a minimally acceptable attempt to solve (see general guidance) their 3TQ This mark can be implied from a correct value of m = 5. If the quadratic is not the correct quadratic, method must be shown.
	A1	For one of $m = 5$ or $n = 16$
	A1	For both $n = 16$ and $m = 5$
ALT1	First B, final 2 A marks – as main scheme	
	M1	For correctly using Pythagoras to form an equation in terms of m and n or x and y. Follow through their CX and CD (ie coming from correct calculation with their p, q, their point D).
	M1	For an unsimplified equation in terms of m and n or x and y correctly using the gradient of l. Follow through their gradient for l, their p and their q. Allow working in p, q or x, y Note this work is sometimes being seen in (b). **As long as it is used in part (d),** the equation found in (b) may be given credit.
	ddM1	For eliminating either m or n from either equation and forming a 3TQ. Allow errors in processing if an initial correct method following from 2 correct unsimplified equations to eliminate one of the variables is seen. Allow working in p, q or x, y Dependent on both previous method marks. Note, the previous M mark is for the unsimplified equation, so if simplified incorrectly, this can be used here to gain this mark (general principle of marking).
	M1	For a minimally acceptable attempt to solve (see general guidance) their 3TQ This mark can be implied from a correct value of m = 5. If the quadratic is not the correct quadratic, method must be shown.

Type MS · Finding a point from a line and a distance

Answer 1 of 10 · Question 6 part d · 7 marks

2024 JUN · Paper 01 · extract 4 of 5 · find part d in this mark scheme extract

ALT2	B1	Any correct unsimplified calculation for CX
	M1	Correct method to find the length of AB
	M1	Correct method to find vector AB
	ddM1	Correctly deduces the relationship between their vectors CX and AB Dependent on both previous method marks.
	M1	Correct method to find the coordinates of X using their vector work.
	A1	For one of $m = 5$ or $n = 16$
	A1	For both $n = 16$ and $m = 5$
ALT3	B1	For a fully correct array as shown oe
	M1	For placing their array (though ft their p and q and their "17") = 80 and correctly multiplying out the discriminant to form an unsimplified equation in m and n
	M1	For an unsimplified equation in terms of m and n or x and y correctly using the gradient of l. Follow through their gradient for l, their p and their q. Allow working in p, q or x, y Note this work is sometimes being seen in (a). **As long as it is used in part (d),** the equation found in (a) may be given credit.
	ddM1	For eliminating either m or n seen either in the array or in their equation formed. Allow errors in processing if an initial correct method following from 2 correct unsimplified equations to eliminate one of the variables is seen. Allow working in p, q or x, y Dependent on both previous method marks. Note, the previous M mark is for the unsimplified equation, so if simplified incorrectly, this can be used here to gain this mark (general principle of marking).
	M1	For solving their linear equation, allow one error in processing, leading to a value of m or n
	A1	For one of $m = 5$ or $n = 16$
	A1	For both $n = 16$ and $m = 5$
If a partial method is seen, with correct answers not from obvious incorrect working, marks may be given (general point of marking unless precluded from the mark scheme for a question). If a partial method is seen and marks cannot be fitted to the main or ALT schemes – send to review.		

Type MS · Finding a point from a line and a distance

Answer 1 of 10 · Question 6 part d · 7 marks

2024 JUN · Paper 01 · extract 5 of 5 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding a point from a line and a distance

Answer 2 of 10 · Question 11 part c · 6 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 4 · find part c in this mark scheme extract

Number		
11 (a)	$\frac{y-6}{x-5}=\frac{6-3}{5--1}\Rightarrow 2y-12=x-5$	M1A1 [2]
(b)	$\left(\frac{2\times5+1\times-1}{2+1},\frac{2\times6+1\times3}{3}\right)\Rightarrow(3,\ 5)*$	M1A1 [2]
(a)		
M1	A fully correct method for finding the equation of a straight line e.g. $\frac{y-y_1}{x-x_1}=\frac{y_2-y_1}{x_2-x_1}$	
A1	$2y-12=x-5$ oe	
(b)		
M1	Use of $\left(\frac{qx_1+px_2}{p+q},\frac{qy_1+py_2}{p+q}\right)$ or $\begin{pmatrix}-1\\3\end{pmatrix}+\frac{2}{3}\times\begin{pmatrix}6\\3\end{pmatrix}$	
A1	Obtains the **given** coordinates	

Type MS · Finding a point from a line and a distance

Answer 2 of 10 · Question 11 part c · 6 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 4 · find part c in this mark scheme extract

(c)	Gradient of perpendicular to AB = -2	B1
	$\frac{5-n}{3-m}=-2\Rightarrow(11=2m+n)$	M1
	Radius = 5, so the length of AC = 10	
	$100=(3-n)^2+(-1-m)^2\Rightarrow(90=n^2-6n+m^2+2m)$	M1
	Solves simultaneous equations $90=121-44m+4m^2-6(11-2m)+m^2+2m$	depM1
	$35=5m^2-30m\Rightarrow m^2-6m-7=0$	depM1
	$(m-7)(m+1)=0\Rightarrow m=7,\ n=-3$	A1 [6]
ALT (c) using vectors		
(c)	$\overrightarrow{AB}=\begin{pmatrix}5\\6\end{pmatrix}-\begin{pmatrix}-1\\3\end{pmatrix}=\begin{pmatrix}6\\3\end{pmatrix}\Rightarrow$ Perpendicular vector to $\overrightarrow{AB}=\begin{pmatrix}3\\-6\end{pmatrix}$	B1
	$\lvert\overrightarrow{AB}\rvert=\sqrt{45}=3\sqrt{5}\Rightarrow\lvert AP\rvert=2\sqrt{5}\qquad\lvert\overrightarrow{AC}\rvert=10$	M1M1
	Using Pythagoras $\lvert\overrightarrow{PC}\rvert=\sqrt{100-20}=\sqrt{80}=4\sqrt{5}$	depM1
	$\overrightarrow{PC}=\frac{4\sqrt{5}}{3\sqrt{5}}\times\begin{pmatrix}3\\-6\end{pmatrix}=\begin{pmatrix}4\\-8\end{pmatrix}$	depM1
	$\Rightarrow$ Coordinates of C are $(3+(4),\ 5+(-8))\Rightarrow(7,-3)$	A1 [6]

(c)

B1 Gradient of perpendicular to AB = -2 (Can be implied by M1)

M1 Obtains an equation using the gradient of the perpendicular and the points (m, n) and P (Condone if given in terms of x and y)

M1 Obtains a second equation using AC = 10 and the points A and C

depM1 Solve simultaneously to obtain a 3 term quadratic

depM1 Solve their 3 term quadratic

A1 $m = 7,\ n = -3$

ALT:

B1 Perpendicular vector to $\overrightarrow{AB}=\begin{pmatrix}3\\-6\end{pmatrix}$

M1 Finds $\lvert\overrightarrow{AB}\rvert$ or $\lvert\overrightarrow{AP}\rvert$ or $\overrightarrow{PC}=\begin{pmatrix}3\\5\end{pmatrix}+\lambda\begin{pmatrix}3\\-6\end{pmatrix}$

M1 States $\lvert\overrightarrow{AC}\rvert=10$ or $\lambda=\frac{4}{3}$

depM1 Use of Pythagoras to find $\lvert\overrightarrow{PC}\rvert$ or $\overrightarrow{PC}=\begin{pmatrix}3\\5\end{pmatrix}+\frac{4}{3}\times\begin{pmatrix}3\\-6\end{pmatrix}$

depM1 Finds $\overrightarrow{PC}$

A1 $m = 7,\ n = -3$

Type MS · Finding a point from a line and a distance

Answer 2 of 10 · Question 11 part c · 6 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 4 · find part c in this mark scheme extract

(d)	$\frac{y-'-3'}{x-'7'}=\frac{1}{2}\Rightarrow\left\{y=\frac{x-13}{2}\right\},\ \frac{y-3}{x--1}=-2\Rightarrow\{y=-2x+1\}$ Solving simultaneous equations by any method $\frac{x-13}{2}=-2x+1\Rightarrow p=3$ and $q=-5$	M1 M1A1 [3]
ALT (d) using vectors		
(d)	$\overrightarrow{AD}=\overrightarrow{PC}=\begin{pmatrix}4\\-8\end{pmatrix}$ Coordinates of point D $\Rightarrow(-1+(4),\ 3+(-8))=(3,-5)$	M1 M1A1 [3]
(e)	Length of AB $\sqrt{(6-3)^2+(5--1)^2}=3\sqrt{5}$ Length of CD $\sqrt{(-3--5)^2+(7-3)^2}=\sqrt{20}=2\sqrt{5}$ Area of trapezium $=\frac{1}{2}(3\sqrt{5}+2\sqrt{5})\times 4\sqrt{5}=50$ (units^2)	M1 A1 M1A1 [4]
ALT (e) using vectors		
	Area $=\frac{1}{2}\begin{vmatrix}-1 & 5 & 7 & 3 & -1\\ 3 & 6 & -3 & -5 & 3\end{vmatrix}$ $=\frac{1}{2}[(-6-15-35+9)-(15+42-9+5)]=50$	 M1A1 M1A1 [4]
		Total 17 marks

Type MS · Finding a point from a line and a distance

Answer 2 of 10 · Question 11 part c · 6 marks

2019 JUN · Paper 01 (regional R) · extract 4 of 4 · find part c in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Finding a point from a line and a distance

Answer 3 of 10 · Question 10 part c · 9 marks

2022 JAN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

10 (a)	For the correct value of $a = 10$	B1 [1]
(b)	Gradient of line L_2 $m = -\frac{1}{2}$ $y - '10' = '-\frac{1}{2}'(x-2)$ $y - 10 = -\frac{1}{2}(x-2)$ oe $x + 2y - 22 = 0$ *	B1 M1 A1 A1 cso [4]
(c)	Coordinates of point A are $(-3,\ 0)$ Coordinates of point B are $(22,\ 0)$ Length of AC $(5\sqrt{2})^2 = (m - '-3')^2 + n^2$ $[\Rightarrow 50 = m^2 + 6m + 9 + n^2]$ Gradient of BC $\frac{1}{4} = \frac{n}{m-'22'} \Rightarrow n = \frac{m-'22'}{4} \Rightarrow n^2 = \frac{m^2-44m+484}{16}$ $50 = m^2 + 6m + 9 + \frac{m^2-44m+484}{16}$ [or $50 = (22+4n)^2 + 6(22+4n) + 9 + n^2$] $17m^2 + 52m - 172 = 0$ OR $17n^2 + 200n + 575 = 0$ e.g. $m = \frac{-52 \pm \sqrt{52^2 - 4\times17\times-172}}{2\times17} \Rightarrow m = 2,\ (-5.058...)$ $m = 2$ and $n = -5$	B1 B1 M1 M1 ddM1 A1 M1 A1A1 [9]
(d)	Area$_{APB} = \frac{1}{2}('10') \times ('22' - -'3') = (125)$ Area$_{ABC} = \frac{1}{2}('5') \times ('22' - '-3') = \left(\frac{125}{2}\right)$ Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2	M1 A1 A1 [3]
	ALT	
	Area$_{ACBP} = \frac{1}{2}('25') \times ('15') = ...$ Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2	[M1A1 A1]
	ALT	

Type MS · Finding a point from a line and a distance

Answer 3 of 10 · Question 10 part c · 9 marks

2022 JAN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

	$A=\frac{1}{2}\begin{bmatrix}\text{'}-3\text{'} & 2 & \text{'}22\text{'} & \text{'}2\text{'} & \text{'}-3\text{'}\\ 0 & \text{'}10\text{'} & 0 & \text{'}-5\text{'} & \text{'}0\text{'}\end{bmatrix}$	[M1
	$A=\frac{1}{2}\left(\left[(-3)\times10+2\times0+22\times(-5)+2\times0\right]-\left[2\times0+22\times10+2\times0+(-3)\times-5\right]\right)=\ldots$	A1
	Area of quadrilateral $APBC = \frac{375}{2}$ (units)^2	A1]
		Total 17 marks

Type MS · Finding a point from a line and a distance

Answer 3 of 10 · Question 10 part c · 9 marks

2022 JAN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

		Any other values of m and n must be rejected.
(d)	**M1**	For either area of triangle APB or ABC $\text{Area}_{APB} = \frac{1}{2}('10')\times('22'--'3') = (125)$ or $\text{Area}_{ABC} = \frac{1}{2}('5')\times('22'-'-3') = \left(\frac{125}{2}\right)$
	A1	Either area of triangle APB or ABC correct
	A1	Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2
ALT		
	M1A1	$\text{Area}_{ACBP} = \frac{1}{2}('25')\times('15') = \ldots$
	A1	Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2
ALT – determinant method		
	M1	For using the 'determinant method' e.g. $A = \frac{1}{2}\begin{bmatrix} '-3' & 2 & '22' & '2' & '-3' \\ 0 & '10' & 0 & '-5' & '0' \end{bmatrix}$
	A1	For a correct evaluation of their determinants using **their** values e.g. $A = \frac{1}{2}\left(\left[(-3)\times10+2\times0+22\times(-5)+2\times0\right]-\left[2\times0+22\times10+2\times0+(-3)\times-5\right]\right) = \ldots$
	A1	Area of quadrilateral $APBC = \frac{375}{2}$ (units)^2

Type MS · Finding a point from a line and a distance

Answer 3 of 10 · Question 10 part c · 9 marks

2022 JAN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding a point from a line and a distance

Answer 4 of 10 · Question 9 part c · 6 marks

2024 JUN · Paper 01 (regional R) · find part c in this mark scheme extract

9(a)	$\frac{y-8}{8-3}=\frac{x-6}{6--4}\Rightarrow\frac{y-8}{5}=\frac{x-6}{10}\Rightarrow\left[y-8=\frac{1}{2}(x-6)\right]$	M1A1 [2]
(b)	$C=\left(\frac{4\times6+1\times-4}{4+1},\frac{4\times8+1\times3}{4+1}\right)=(4,7)$	B1,B1 [2]
(c)	$-2=\frac{q-7}{p-4}\Rightarrow q=15-2p$ $8\sqrt{5}=\sqrt{(4-p)^2+(7-q)^2}$ $\Rightarrow 320=(4-p)^2+(2p-8)^2\Rightarrow p^2-8p-48=0$ $\Rightarrow(p+4)(p-12)=0$ $\Rightarrow p=-4\quad q=15-2\times-4=23\Rightarrow D=(-4,23)$	M1 M1 M1 M1 A1A1 [6]
(d)	Area of triangle $ACD=\frac{1}{2}\times(23-3)\times(4--4)=80$ [square units] **ALT 1** $\frac{1}{2}\begin{bmatrix}-4 & 4 & -4 & -4\\ 3 & 7 & 23 & 3\end{bmatrix}=\frac{1}{2}\left[(-28+92-12)-(12-28-92)\right]=80$ **ALT 2** $A=\frac{1}{2}\times AC\times CD=\frac{1}{2}\times4\sqrt{5}\times8\sqrt{5}=80$	M1A1 [2] [M1A1] [M1A1]
		Total 12 marks

Type MS · Finding a point from a line and a distance

Answer 5 of 10 · Question 9 part b · 8 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

Number		
9 (a)	$a=-1,\ b=-2$	B1,B1 [2]
(b)	Gradient of $l_1=-2,\Rightarrow$ Gradient of $l_2=\frac{1}{2}$	B1,B1
	$180=(x+1)^2+(y-6)^2$	M1
	$\frac{1}{2}=\frac{y-6}{x+1}\Rightarrow x=2y-13$	M1
	Solves simultaneous equations;	
	$180=([2y-13]+1)^2+(y-6)^2\Rightarrow 0=5y^2-60y$	M1M1
	or $180=(x+1)^2+\left(\frac{1}{2}x+\frac{13}{2}-6\right)^2\Rightarrow 0=x^2+2x-143=0$	
	$y=0,\ y=12\Rightarrow x=-13,\ x=11$ or $x=-13,\ x=11\Rightarrow y=0,\ y=12$	A1A1
	Coordinates are $(-13,\ 0)$ and $(11,\ 12)$	[8]
(c)	Area of triangle PQR	
	$PQ=\sqrt{(6+2)^2+(-1-3)^2}=4\sqrt{5}$	M1
	Area $=\frac{1}{2}\times 4\sqrt{5}\times 6\sqrt{5}=60$ (units)^2	M1A1 [3]
	ALT	
	Area $=\frac{1}{2}\begin{pmatrix}-13 & -1 & 3 & -13\\ 0 & 6 & -2 & 0\end{pmatrix}=\frac{1}{2}(-78+2+0-0-18-26)=-60$	{M1}{M1}
	$\Rightarrow 60$ (units)^2	{A1} [3]
(d)	Coordinates of R required are $(-13,\ 0)$	
	$\angle RPQ=90^\circ$ so RQ is a diameter	
	$\left(\frac{-13+3}{2},\frac{0-2}{2}\right)\Rightarrow(-5,\ -1)$	M1A1 [2]
		Total 15 marks

Type MS · Finding a point from a line and a distance

Answer 5 of 10 · Question 9 part b · 8 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

(b)	
B1	Gradient of $l_1 = -2$
B1	Gradient of $l_2 = \frac{1}{2}$
M1	Use of $PR = 6\sqrt{5}$ to obtain an equation
M1	Use of gradient of the perpendicular to obtain an equation
M1	Solves simultaneously
M1	Simplifies to $5y^2 - 60y = 0$ or $x^2 + 2x - 143 = 0$
A1	All 4 values identified 0, 12, –13, 11
A1	$(11,\ 12)$ and $(-13,\ 0)$ (must be paired correctly and if written as a coordinate then must be in the correct order)
(c)	
M1	$PQ = 4\sqrt{5}$
M1	Use of Area $= \frac{1}{2} \times PQ \times PR$
A1	60 (units)^2
ALT	
M1	Use of Area $= \frac{1}{2}\begin{pmatrix} -13 & -1 & 3 & -13 \\ 0 & 6 & -2 & 0 \end{pmatrix}$ ft R provided $e < 0$
M1	$\frac{1}{2}(-78 + 2 + 0 - 0 - 18 - 26)$ ft R provided $e < 0$
A1	60 (units)^2
(d)	
M1	$\left(\frac{-13+3}{2}, \frac{0-2}{2}\right)$ ft R provided $e < 0$
A1	$(-5,\ -1)$

Type MS · Finding a point from a line and a distance

Answer 5 of 10 · Question 9 part b · 8 marks

2020 JAN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding a point from a line and a distance

Answer 6 of 10 · Question 11 part b · 6 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 5 · find part b in this mark scheme extract

11 (a)	$b = 33$	B1
	$\left(\left(\frac{3\times0+2\times10}{5}, \frac{3\times3+2\times33}{5}\right)\right) = (4, 15)$	B1 B1
	ALT	
	Identifies the correct x coordinate of $x = 4$ using any method.	B1
	$y = 3\times'4'+3 = ... \Rightarrow y = 15$	B1B1
	For the gradient of line k $\quad m = -\frac{1}{"3"}$	M1
	$y - "15" = -\frac{1}{"3"}(x - "4")$	dM1
	$3y + x - 49 = 0^*$ cso	A1* cso [6]
(b)	$A = (0, 3)$	B1
	$-\frac{1}{"3"} = \frac{q-"3"}{p(-0)} (\Rightarrow p = -"3"(q-"3"))$ **alt** $y = -\frac{1}{3}x + 3 \Rightarrow q = -\frac{p}{3} + 3$ **OR** $p^2 + (q-"3")^2 = 1440$	M1
	Equation in *q* $10q^2 - 60q + 90 = 1440 \Rightarrow q^2 - 6q - 135 = 0$ oe $10(q-3)^2 = 1440$ or $q^2 - 6q - 135 = 0$	dM1
	Equation in *p* $1440 = p^2 + \left(3 - 3 - \frac{p}{3}\right)^2 \Rightarrow 1440 = \frac{10p^2}{9}$ And attempting to solve by any valid method, reaching a value for p	
	$q - 3 = \pm\sqrt{144}$ or $(q-15)(q+9)$ or $\frac{--6 \pm \sqrt{(-6)^2 - 4.(1).(-135)}}{2\times1}$	ddM1
	$q = 15, p = -36$	A1 A1 [6]
	ALT	
	$A = (0, 3)$	[B1
	$a\sqrt{(-3)^2 + 1^2}$ $(= 12\sqrt{10})$	M1
	$\Rightarrow a = (12)$	dM1
	$p = 0 - 3\times"12"$	dM1
	$q = 3 + 1\times"12"$	
	$p = -36$ and $q = 15$ or D is $(-36, 15)$	A1 A1]

Type MS · Finding a point from a line and a distance

Answer 6 of 10 · Question 11 part b · 6 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 5 · find part b in this mark scheme extract

(c)	**Finding the coordinates of points *E* and *F***	
	(Finding E $x = -36$ $3y + x - 49 = 0$)	M1
	$3y + ("-36") - 49 = 0 \left(y = \frac{85}{3} \right)$	
	(Finding F $x = -36$ $y - 3x = 3$)	M1
	$y - 3("-36") = 3$ $(y = -105)$	
	Find the area of triangle *ECF*	
	Method 1	
	$(CE =) \sqrt{("4" - "-36")^2 + \left("\frac{85}{3}" - "15" \right)^2} = \left(\frac{40\sqrt{10}}{3} \right)$	M1
	Or	
	$(CF =) \sqrt{("4" - "-36")^2 + ("15" - "-105")^2} = \left(40\sqrt{10} \right)$	
	$\left(\text{Area triangle } ECF = \right) \frac{1}{2} \times \frac{40\sqrt{10}}{3} \times 40\sqrt{10} = \frac{8000}{3}$ oe	M1A1 [5]
	Method 2	
	$\text{Area} = \frac{1}{2} \begin{bmatrix} "-36" & "4" & "-36" & "-36" \\ "\frac{85}{3}" & "15" & "-105" & "\frac{85}{3}" \end{bmatrix}$ oe	M1
	$\frac{1}{2} \left(\begin{bmatrix} "-36" \times "15" + "4" \times "-105" + "-36" \times "\frac{85}{3}" \end{bmatrix} - \begin{bmatrix} "-36" \times "-105" + "-36" \times "15" + + "4" \times "\frac{85}{3}" \end{bmatrix} \right) = \left(\frac{10060}{3} - [-1980] \right) = \frac{8000}{3}$ oe	M1A1
	Method 3	
	Finds the lengths: $DC = 4 - (-36) = 40$ and $EF = \frac{85}{3} - (-105) = \frac{400}{3}$	M1
	$\text{Area} = \frac{1}{2} \times '40' \times '\frac{400}{3}' = \frac{8000}{3}$ oe	M1A1
		Total 17 marks

Type MS · Finding a point from a line and a distance

Answer 6 of 10 · Question 11 part b · 6 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 5 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	Correct value of b. This may be embedded in a coordinate of B (10, 33)
	B1	For either $x = 4$ or $y = 15$
	B1	For (4, 15) or both $x = 4$ and $y = 15$
	ALT (Without finding the value of b)	
	B1	For $x = 4$ [by any method]
	B1	Substitutes their value of $x = 4$ into the given equation of l $y = 3\times'4'+3 = ...$
	B1	For $y = 15$
	M1	For identification of a gradient from the equation of l and use of $m = -\frac{1}{"3"}$
	dM1	For the correct use of any form of the equation of a line, if using $y = mx + c$, must reach $c = ...$ Note that $c = \frac{49}{3}$ Use of their (4, 15) which **must** have come from an attempt to find the midpoint of AB and their m. **Note:** Dependent upon the previous method mark.
	A1	$3y + x - 49 = 0$ accept terms on one side in any order. Cso **Note:** This equation is given. Award this and the previous mark that has come from work without any errors or omissions.
(b)	B1	Identifies the y intercept of line l (0, 3). This can be implied by later correct work or accept seen in part (a) Look for this coordinate written on a sketch if it is not in the body of the work.
	M1	Equates $-\frac{1}{"3"}$ to a **correct** expression for the gradient in terms of p and q, using **their** values of (0, 3) for the y intercept. **OR** for **correct** use of Pythagoras for the length of AD to form an equation in p and q, using **their** value of (0, 3) for the y intercept.
	dM1	For forming a quadratic equation in either p or q which must come from using their expression for the gradient and the length. **This is dependent on the first M mark in (b)**
	ddM1	For solving their quadratic equation by any acceptable method. **This is dependent on the previous two M marks in (b)**
	A1	For a correct p **or** q.
	A1	For a correct p **and** q.
	ALT	
	B1	Identifies the y intercept of line l (0, 3) This can be implied by later correct work or accept seen in part (a) Look for this coordinate written on a sketch if it is not in the body of the work.
	M1	Uses a correct method, scaling the gradient of $-\frac{1}{3}$ Must arrive at a scaling value (shown as a in the mark scheme), but it may not be the correct value. This mark may be implicit in work that follows.
	dM1	Uses their scaling factor, with the coordinates of A, to find a value for p or q **This is dependent on the first M mark in (b)**
	dM1	Uses their scaling factor, with the coordinates of A, to find a value for p and q **This is dependent on the first M mark in (b)**
	A1 A1	First A1 for p **or** q Second A1 for p **and** q
(c)	M1	Substitutes their value for – 36 into the equation for line k to find the y coordinate of point E
	M1	Substitutes their value for – 36 into the equation for line l to find the y coordinate of point F

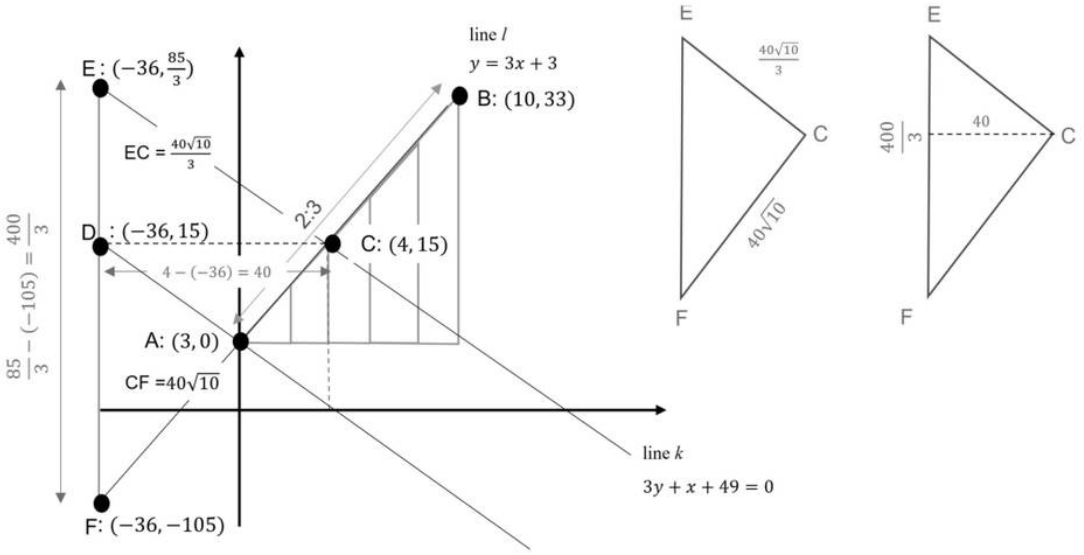
Type MS · Finding a point from a line and a distance

Answer 6 of 10 · Question 11 part b · 6 marks

2022 JUN · Paper 01 (regional R) · extract 4 of 5 · find part b in this mark scheme extract

	Method 1	
	M1	Uses Pythagoras with **their** values to find the length of CE **and** CF correctly.
	M1	Correct method to find the area of the triangle, using **their** values.
	A1	For $\frac{8000}{3}$ oe
	Method 2	
	M1	For a correct array using their values.
	M1	Correct calculation shown, using their values, takes the modulus and multiplies by $\frac{1}{2}$
	A1	$\frac{8000}{3}$ oe
	Method 3	
	M1	For finding the lengths DC **and** EF correctly using **their** values.
	M1	Applies the correct formula for the area of a triangle using their values.
	A1	For $\frac{8000}{3}$ oe

USEFUL SKETCH



Type MS · Finding a point from a line and a distance

Answer 6 of 10 · Question 11 part b · 6 marks

2022 JUN · Paper 01 (regional R) · extract 5 of 5 · find part b in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Finding a point from a line and a distance

Answer 7 of 10 · Question 10 part d · 6 marks

2025 NOV · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

Question	Scheme	Marks
10 (a)	$10-3a+a=0 \Rightarrow 10=2a \Rightarrow a=5$ *	M1A1cso [2]
(b)	$(0,-5)$	B1 [1]
(c) (i)	$A \quad (5,10)$ $C \quad (50,-5)$ $y=3x-5 \Rightarrow m_{L1}=3 \qquad m_{L2}=\dfrac{-5-10}{50-5}=\dfrac{-1}{3}$ $m_{L2}\times m_{L1}=\dfrac{-1}{3}\times 3=-1$	B1 M1A1 [3]
(ii)	$y-(-5)=\dfrac{-1}{3}(x-50) \Rightarrow y=\dfrac{-x}{3}+\dfrac{35}{3}$	M1A1 [2]
(d)	$-3=\dfrac{n-("-5")}{m} \Rightarrow n=-5-3m$ $2250=(50-m)^2+(-5-n)^2$ $2250=(50-m)^2+(-5-(-5-3m))^2=(50-m)^2+(3m)^2$ $2250=2500-100m+10m^2 \Rightarrow 0=25-10m+m^2=(m-5)^2$ $\Rightarrow m=5$ $\Rightarrow n=-5-3\times 5=-20$ **ALT** $-3=\dfrac{n-("-5")}{m} \Rightarrow m=\dfrac{-(n+5)}{3}$ $2250=(50-m)^2+(-5-n)^2$ $2250=\left(50+\dfrac{n+5}{3}\right)^2+n^2+10n+25$ $2250=2500+\dfrac{100(n+5)}{3}+\dfrac{n^2+10n+25}{9}+n^2+10n+25$ $\Rightarrow \quad 0=9(2500-2250)+300(n+5)+10n^2+100n+250$ $=2500+300n+1500+10n^2+100n$ $=n^2+40n+400$ $=(n+20)(n+20)$ $\Rightarrow n=-20$ $\Rightarrow m=\dfrac{-(-20+5)}{3}=5$	M1 M1 M1 M1 A1 A1 [6] [M1] [M1] [M1] [M1] [A1] [A1]
(e)	Correctly uses the formula for a kite using their lengths	M1

Type MS · Finding a point from a line and a distance

Answer 7 of 10 · Question 10 part d · 6 marks

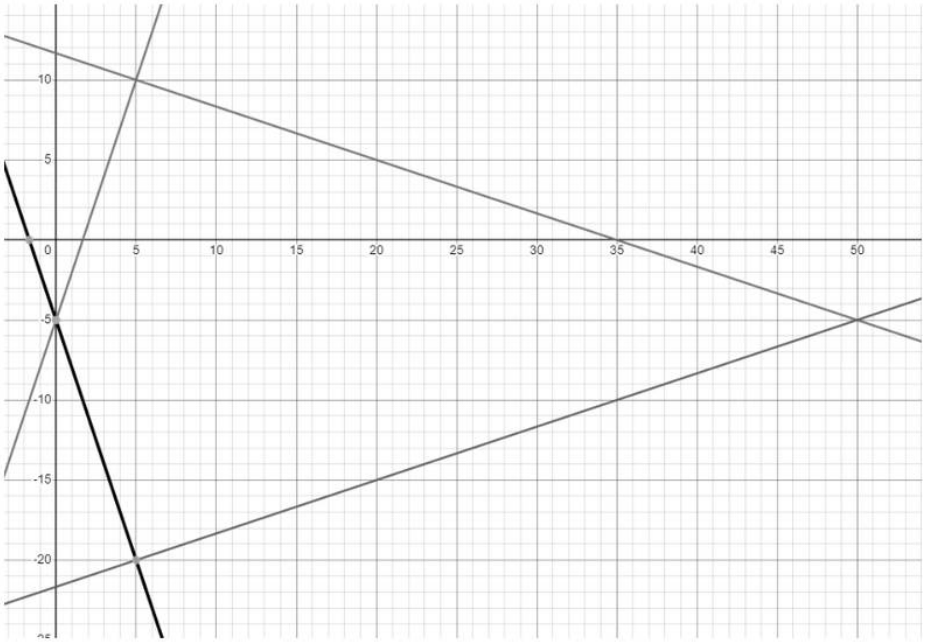
2025 NOV · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

	$\text{Area} = \frac{50\times(10-(-20))}{2} = 750$	dM1A1 [3]
	ALT 1 Correctly uses determinant method with their co-ordinates $\text{Area} = \frac{1}{2}\begin{pmatrix} 0 & 5 & 50 & 5 & 0 \\ -5 & 10 & -5 & -20 & -5 \end{pmatrix} = 750$	[M1 dM1A1]
	ALT 2 Correctly uses formulae for two triangles ABD and ACD using their lengths $\text{Area} = 2\times\left(\frac{1}{2}\times 15\times 50\right) = 750$ Or $\text{Area} = \left(\frac{1}{2}\times 30\times 5\right)+\left(\frac{1}{2}\times 45\times 30\right) = 750$	[M1 dM1A1]
		Total 17 marks

Type MS · Finding a point from a line and a distance

Answer 7 of 10 · Question 10 part d · 6 marks

2025 NOV · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract



Type MS · Finding a point from a line and a distance

Answer 7 of 10 · Question 10 part d · 6 marks

2025 NOV · Paper 01 · extract 4 of 4 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding a point from a line and a distance

Answer 8 of 10 · Question 4 part c · 6 marks

2021 JUN · Paper 02 · extract 1 of 5 · find part c in this mark scheme extract

number		
4 (a)	Gradient $= \dfrac{11+10}{3+4} = 3$	
	$y+10=3(x+4)$ or $y-11=3(x-3)$ oe	M1A1 (2)
(b)	e.g. $\left(\dfrac{4\times-4+3\times3}{3+4}, \dfrac{4\times-10+3\times11}{3+4}\right)=(-1,-1)$	M1 A1 (2)
ALT (b)	Using Vectors	
	$\begin{pmatrix}-4\\-10\end{pmatrix}+\dfrac{3}{7}\begin{pmatrix}7\\21\end{pmatrix}=\begin{pmatrix}-1\\-1\end{pmatrix}$ or $\begin{pmatrix}3\\11\end{pmatrix}-\dfrac{4}{7}\begin{pmatrix}7\\21\end{pmatrix}=\begin{pmatrix}-1\\-1\end{pmatrix}$	{M1} {A1}
(c)	$-\dfrac{1}{3}=\dfrac{n+1}{m+1} \Rightarrow -\dfrac{1}{3}(m+1)=n+1$	M1
	$\left(\sqrt{10}\right)^2=(m+1)^2+(n+1)^2$	M1
	$10=(m+1)^2+\dfrac{1}{9}(m+1)^2$	M1
	$9=(m+1)^2$	M1
	$m=-4 \quad n=0$	A1 A1 (6)
ALT (c)	Using Vectors	
	$\overrightarrow{AB}=\begin{pmatrix}7\\21\end{pmatrix}$ so perpendicular to $\overrightarrow{AB}=\begin{pmatrix}21\\-7\end{pmatrix}$	{M1}
	$\left\lvert\overrightarrow{AB}\right\rvert=7\sqrt{10} \Rightarrow, \left\lvert\overrightarrow{AP}\right\rvert=3\sqrt{10}$	{M1,M1}
	$\overrightarrow{PQ}=\dfrac{\sqrt{10}}{7\sqrt{10}}\times\begin{pmatrix}21\\-7\end{pmatrix}=\begin{pmatrix}3\\-1\end{pmatrix}$	{M1}
	So $Q=(-1-3,-1--1)$	{A1}
	$Q=(-4,0)$	{A1}
(d)(i)	$AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$	M1
	$RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$	A1
(d)(ii)	Gradient of $RQ=\dfrac{-21-0}{-11+4}=3$	M1
	So Gradient of AB (=3) = Gradient of RQ	A1 (4)

Type MS · Finding a point from a line and a distance

Answer 8 of 10 · Question 4 part c · 6 marks

2021 JUN · Paper 02 · extract 2 of 5 · find part c in this mark scheme extract

ALT (d)	Using Vectors $\overrightarrow{RQ} = \begin{pmatrix} -4-(-11) \\ 0-(-21) \end{pmatrix} = \begin{pmatrix} 7 \\ 21 \end{pmatrix}$	{M1} {A1}
	$\overrightarrow{AB} = \begin{pmatrix} 7 \\ 21 \end{pmatrix} = \overrightarrow{RQ}$	{M1}
	Because the vectors are the same they must be parallel and the same length	{A1}
(e)	Area = $7\sqrt{10} \times \sqrt{10} = 70$	M1 A1 (2)
ALT (e)	Using Vectors $\frac{1}{2}\begin{vmatrix} 3 & -4 & -11 & -4 & 3 \\ 11 & 10 & -21 & 0 & 11 \end{vmatrix}$	{M1}
	$= 70$	{A1}
		Total is 16 marks

Type MS · Finding a point from a line and a distance

Answer 8 of 10 · Question 4 part c · 6 marks

2021 JUN · Paper 02 · extract 3 of 5 · find part c in this mark scheme extract

Part	Mark	Guidance
(a)	M1	For a fully correct method of finding an equation of a straight line. $\frac{y-y_1}{y_2-y_1}=\frac{x-x_1}{x_2-x_1}\Rightarrow\frac{y-(-10)}{11-(-10)}=\frac{x-(-4)}{3-(-4)}$ Or finds gradient $\frac{11+10}{3+4}=3$ and uses $y+10=3(x+4)$ or $y-11=3(x-3)$ If $y=mx+c$ is used, they must find a complete equation for this mark. Allow **one** error only for the award of this mark.
	A1	For a correct line in any form. $y+10=3(x+4)$ or $y-11=3(x-3)$ or $y=3x+2$ or even $\frac{y+10}{21}=\frac{x+4}{7}$ but do not allow incomplete processing.
(b)	M1	For one correct from $x=-1$ or $y=-1$
	A1	For the correct coordinates of point P $(-1,-1)$ Accept $x=-1$ $y=-1$
(c)	M1	Uses the perpendicular gradient to set up an equation in m and n. $-\frac{1}{'3'}=\frac{n-'(-1)'}{m-'(-1)'}\Rightarrow-\frac{1}{'3'}(m+1)=n+1$ or $n=-\frac{1}{3}m-\frac{4}{3}$ Ft their gradient in part (a) and their P from part (b) for this mark.
	M1	Uses Pythagoras theorem to set up an equation in m and n. $\left(\sqrt{10}\right)^2=(m-'(-1)')^2+(n-'(-1)')^2$ Ft their coordinates of point P form part (b) for this mark.
	M1	Attempts to solve their two equations in n and m simultaneously and forms a quadratic equation in one variable only. $10=(m+1)^2+\frac{1}{9}(m+1)^2\Rightarrow 9=(m+1)^2$ or $0=m^2+2m-8$ or $10=9(n+1)^2+(n+1)^2\Rightarrow 0=10n^2+20n$
	M1	For solving their either: $9=(m+1)^2\Rightarrow m=...$ or $0=10n^2+20n\Rightarrow n=...$ which must be a quadratic equation.
	A1	For finding **either** $m=-4$ **or** $n=0$ Condone the sight of $m=2$ for this mark.
	A1	For finding **both** $m=-4$ **and** $n=0\Rightarrow(-4,\ 0)$ The final answer must be given as coordinates.
	ALT – using vectors – see main scheme.	
(d)(i)	M1	For finding **either** the length $AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$ **Or** $RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$
	A1	For finding **both** the length $AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$ **And** $RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$ **and** states they are equal
(d)(ii)	M1	The gradient of $RQ=\frac{-21-'0'}{-11-'(-4)'}='3'$ Ft their coordinates from part (c)

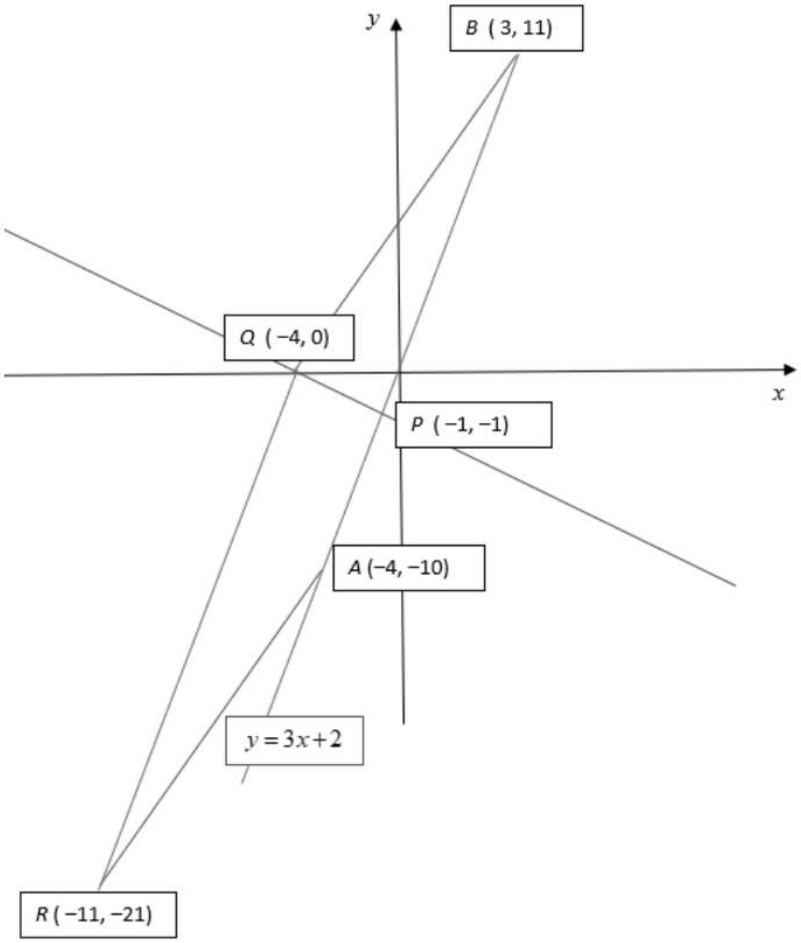
Type MS · Finding a point from a line and a distance

Answer 8 of 10 · Question 4 part c · 6 marks

2021 JUN · Paper 02 · extract 4 of 5 · find part c in this mark scheme extract

	A1	States that the gradient of RQ = gradient of AB [from (a)]
	ALT – Uses vectors, see main scheme. Ft their coordinates of Q – (m, n)	
(e)	**M1**	For a correct expression for the area using their length of AB and the given length of PQ $\left(\sqrt{10}\right)$ Area = $'7\sqrt{10}'\times\sqrt{10}=...$
	A1	For the area = 70 [square units]
	ALT – Uses the discriminant	
	M1	For a correct expression of the area in sequential order using their coordinates for Q Area = $\frac{1}{2}\begin{vmatrix} 3 & -4 & -11 & '-4' & 3 \\ 11 & 10 & -21 & '0' & 11 \end{vmatrix}$
	A1	Area = 70 [square units]

Useful sketch



Type MS · Finding a point from a line and a distance

Answer 8 of 10 · Question 4 part c · 6 marks

2021 JUN · Paper 02 · extract 5 of 5 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding a point from a line and a distance

Answer 9 of 10 · Question 8 part d.i · 0 marks

2023 JUN · Paper 02 · extract 1 of 5 · find part d.i in this mark scheme extract

Question	Scheme	Marks
8(a)	$(\text{Grad}_{AB}) = \frac{2-8}{12-(-6)}$ oe $y-2="\left(-\frac{1}{3}\right)"(x-12)$ oe $\Rightarrow x+3y-18=0$ or $-x-3y+18=0$ oe	M1 dM1 (A1 on ePen) A1 [3]
(b)	Length $=\sqrt{(12--6)^2+(2-8)^2}=6\sqrt{10}$ eg $\sqrt{360}$ oe	M1A1 [2]
(c)	$\left(\left(\frac{2\times-6+1\times12}{1+2}, \frac{2\times8+1\times2}{1+2}\right)\right)$ $(0,6)$ or $m=0, n=6$	B1 B1 (M1A1 on ePen) [2]
(d) Mark parts (i) and (ii) together	Gradient of $CA=\frac{q-8}{p+6}$ **or** gradient of $CB=\frac{q-2}{p-12}$ oe $\frac{q-8}{p+6}=-\frac{1}{\frac{q-2}{p-12}}=-\left(\frac{p-12}{q-2}\right)\Rightarrow q^2-10q+16=-p^2+6p+72$ Gradient of $XC=-\frac{1}{"-\frac{1}{3}"}\Rightarrow-\frac{1}{"-\frac{1}{3}"}=\frac{q-6}{p-0}(\Rightarrow q=3p+6)$ $('3p+6')^2-10('3p+6')+16=-p^2+6p+72\Rightarrow 10p^2-80=0$ $10p^2-80=0\Rightarrow p=\sqrt{8}$ oe $q=3\times\sqrt{8}+6=6+6\sqrt{2}$ oe	M1 M1 B1ft ddM1A1 M1A1 [7]
ALT1 Mark parts (i) and (ii) together	$(\text{midpoint of } AB=)\left(\frac{-6+12}{2}, \frac{8+2}{2}\right)$ $(=(3,5))$ $(\text{radius of } C)=\frac{"6\sqrt{10}"}{2}$ oe or $\sqrt{(-6-3)^2+(8-5)^2}$ oe $y=\frac{-1}{"-\frac{1}{3}"}x+6$ $(y=3x+6)$ oe $(x-"3")^2+(y-"5")^2=\left(\frac{"6\sqrt{10}"}{2}\right)^2\Rightarrow(x-"3")^2+("3x+6"-"5")^2=\left(\frac{"6\sqrt{10}"}{2}\right)^2$ $10x^2=80$ oe $x=\sqrt{8}$ oe $p=\sqrt{8}, q=6+3\sqrt{8}$ oe	M1 M1 B1ft ddM1 A1 M1 A1 [7]

Type MS · Finding a point from a line and a distance

Answer 9 of 10 · Question 8 part d.i · 0 marks

2023 JUN · Paper 02 · extract 2 of 5 · find part d.i in this mark scheme extract

ALT2 Mark parts (i) and (ii) together	$\left((AC)^2=\right)(p--6)^2+(q-8)^2$ or $(AC=)\sqrt{(p--6)^2+(q-8)^2}$ oe	M1
	$\left((BC)^2=\right)(p-12)^2+(q-2)^2$ or $(AC=)\sqrt{(p-12)^2+(q-2)^2}$ oe	M1
	$q=3p+6$	B1ft
	$\left((AB)^2=(AC)^2+(BC)^2\right)$	
	$\left("6\sqrt{10}"\right)^2=(p--6)^2+("3p+6"-8)^2+(p-12)^2+("3p+6"-2)^2$	ddM1A1
	$10p^2=80$ oe	M1
	$p=\sqrt{8}$ oe	A1
	$p=\sqrt{8}, q=6+3\sqrt{8}$ oe	[7]
(e)	Length $CX=\sqrt{\left("\sqrt{8}"-"0"\right)^2+\left("6+6\sqrt{2}"-"6"\right)^2}\left(=4\sqrt{5}\right)$	M1
	Area of triangle $ABC=\frac{1}{2}\times"4\sqrt{5}"\times"6\sqrt{10}"=60\sqrt{2}$ oe	dM1 A1 [3]
ALT	$\frac{1}{2}\begin{vmatrix}-6 & 12 & "\sqrt{8}" & -6\\ 8 & 2 & "6+3\sqrt{8}" & 8\end{vmatrix}$ oe	M1
	$\frac{1}{2}\left[\left(-6\times2+12\times"(6+3\sqrt{8})"+"\sqrt{8}"\times8\right)-\left(-6\times"(6+3\sqrt{8})"+"\sqrt{8}"\times2+12\times8\right)\right]$ oe	dM1
	$60\sqrt{2}$ oe	A1 [3]
		Total 17 marks

Type MS · Finding a point from a line and a distance

Answer 9 of 10 · Question 8 part d.i · 0 marks

2023 JUN · Paper 02 · extract 3 of 5 · find part d.i in this mark scheme extract

Part	Mark	Notes
(a)	M1	For correctly finding the gradient of AB in unsimplified form.
	dM1 (A1 on ePen)	For a full and correct attempt to find the equation of the line using their gradient. No simplification is required. If using $y = mx + c$, a value for c must be found.
	A1	For a correct equation in the required form.
(b)	M1	For using a correct method to find the length of line segment AB, in unsimplified form.
	A1	For the correct exact length.
(c)	B1 (M1 on ePen)	For either coordinate correct.
	B1 (A1 on ePen)	For both correct coordinates (0, 6) For part c the values of m and n can be explicitly identified or written in a coordinate.
In part (d) allow p to be interchangeable with x, q to be interchangeable with y throughout		
(d)	M1	For a correct statement of the gradient for either CA or CB
	M1	For using the negative perpendicular of either gradient and equating the gradients to form an equation in terms of p and q only.
	B1ft	For finding the negative reciprocal of their gradient of XC and placing this equal to a correct expression in p and q as shown.
	ddM1	For correctly substituting their **linear** expression for p or q into a **quadratic** equation in q or p to obtain an equation in one variable. Must use their $q = 3p + 6$ and is dependent on both previous method marks.
	A1	For the correct two term quadratic..
	M1	For **correctly solving their** quadratic to find a value for either p or q
	A1	For **both** p and q correct.
ALT1 (d)	M1	For the correct method to find the midpoint of AB
	M1	For the correct method to find the radius of C, ft their answer from part b if used.
	B1ft	For the equation of the line, unsimplified, ft their gradient of AB
	ddM1	For correctly substituting their $y = 3x + 6$ into the equation of a circle, using their midpoint of AB and their radius of C Must use their $y = 3x + 6$ and is dependent on both previous method marks.
	A1	Correct equation
	M1	For **correctly solving their** quadratic to find a value for either x or y
	A1	For **both** p and q correct.
ALT2 (d)	M1	For the correct method to find the length of AC or $(AC)^2$
	M1	For the correct method to find the length of BC or $(BC)^2$
	B1ft	For the equation of the line, unsimplified, ft their gradient of AB
	ddM1	For correctly substituting their lengths and their $q = 3p + 6$ into a correct Pythagorean equation, dependent on both previous method marks.
	A1	Correct equation
	M1	For **correctly solving their** quadratic to find a value for either p or q
	A1	For **both** p and q correct.

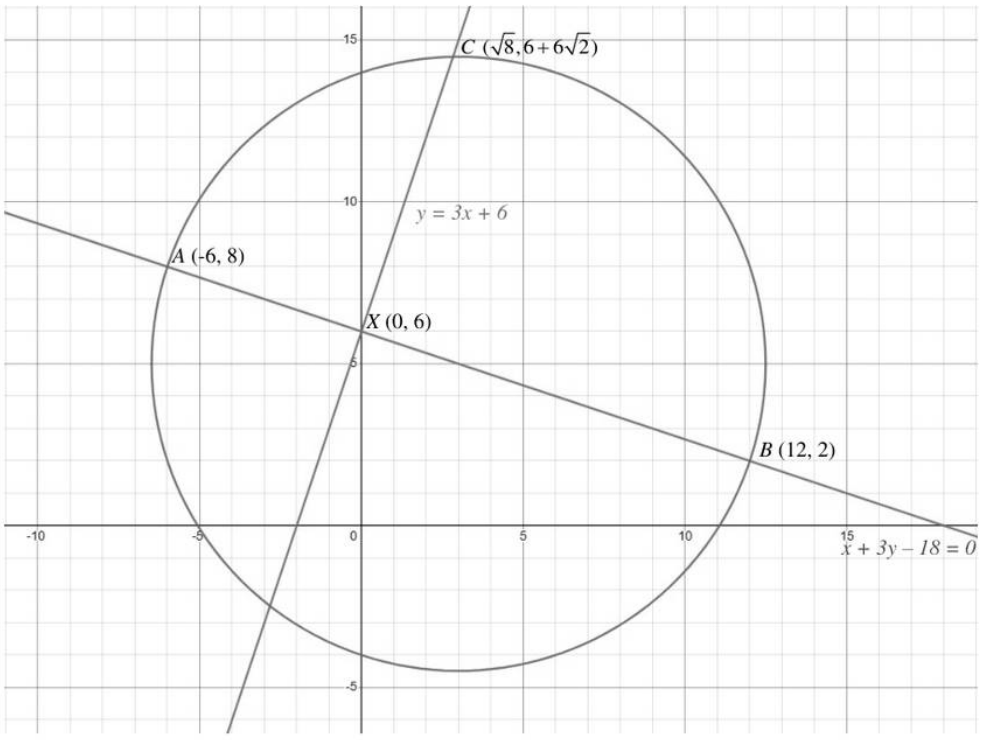
Type MS · Finding a point from a line and a distance

Answer 9 of 10 · Question 8 part d.i · 0 marks

2023 JUN · Paper 02 · extract 4 of 5 · find part d.i in this mark scheme extract

(e)	**M1**	For using a correct method to find the length of the perpendicular from AB to C
	dM1	For using their results from part (b) and their length of perpendicular from AB to C with the correct formula for the area of a triangle. Dependent on previous method mark.
	A1	For the correct area of $60\sqrt{2}$ or $30\sqrt{8}$ (units^2)
ALT	**M1**	For a correct statement for the area such as the one shown, using their values of p and q
	dM1	For the correct evaluation of their determinant Dependent on previous method mark.
	A1	For the correct area of $60\sqrt{2}$ or $30\sqrt{8}$ (units^2)

Useful Sketch for Parts c/d – look for any working on or near a sketch.



Type MS · Finding a point from a line and a distance

Answer 9 of 10 · Question 8 part d.i · 0 marks

2023 JUN · Paper 02 · extract 5 of 5 · find part d.i in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding a point from a line and a distance

Answer 10 of 10 · Question 8 part d.ii · 7 marks

2023 JUN · Paper 02 · extract 1 of 5 · find part d.ii in this mark scheme extract

Question	Scheme	Marks
8(a)	$(\text{Grad}_{AB}) = \frac{2-8}{12-(-6)}$ oe $y-2="\left(-\frac{1}{3}\right)"(x-12)$ oe $\Rightarrow x+3y-18=0$ or $-x-3y+18=0$ oe	M1 dM1 (A1 on ePen) A1 [3]
(b)	Length $=\sqrt{(12--6)^2+(2-8)^2}=6\sqrt{10}$ eg $\sqrt{360}$ oe	M1A1 [2]
(c)	$\left(\left(\frac{2\times-6+1\times12}{1+2}, \frac{2\times8+1\times2}{1+2}\right)\right)$ $(0,6)$ or $m=0, n=6$	B1 B1 (M1A1 on ePen) [2]
(d) Mark parts (i) and (ii) together	Gradient of $CA=\frac{q-8}{p+6}$ **or** gradient of $CB=\frac{q-2}{p-12}$ oe $\frac{q-8}{p+6}=-\frac{1}{\frac{q-2}{p-12}}=-\left(\frac{p-12}{q-2}\right)\Rightarrow q^2-10q+16=-p^2+6p+72$ Gradient of $XC=-\frac{1}{"-\frac{1}{3}"}\Rightarrow-\frac{1}{"-\frac{1}{3}"}=\frac{q-6}{p-0}(\Rightarrow q=3p+6)$ $('3p+6')^2-10('3p+6')+16=-p^2+6p+72\Rightarrow 10p^2-80=0$ $10p^2-80=0\Rightarrow p=\sqrt{8}$ oe $q=3\times\sqrt{8}+6=6+6\sqrt{2}$ oe	M1 M1 B1ft ddM1A1 M1A1 [7]
ALT1 Mark parts (i) and (ii) together	$(\text{midpoint of } AB=)\left(\frac{-6+12}{2}, \frac{8+2}{2}\right)$ $(=(3,5))$ $(\text{radius of } C)=\frac{"6\sqrt{10}"}{2}$ oe or $\sqrt{(-6-3)^2+(8-5)^2}$ oe $y=\frac{-1}{"-\frac{1}{3}"}x+6$ $(y=3x+6)$ oe $(x-"3")^2+(y-"5")^2=\left(\frac{"6\sqrt{10}"}{2}\right)^2\Rightarrow(x-"3")^2+("3x+6"-"5")^2=\left(\frac{"6\sqrt{10}"}{2}\right)^2$ $10x^2=80$ oe $x=\sqrt{8}$ oe $p=\sqrt{8}, q=6+3\sqrt{8}$ oe	M1 M1 B1ft ddM1 A1 M1 A1 [7]

Type MS · Finding a point from a line and a distance

Answer 10 of 10 · Question 8 part d.ii · 7 marks

2023 JUN · Paper 02 · extract 2 of 5 · find part d.ii in this mark scheme extract

ALT2 Mark parts (i) and (ii) together	$\left((AC)^2=\right)(p--6)^2+(q-8)^2$ or $(AC=)\sqrt{(p--6)^2+(q-8)^2}$ oe	M1
	$\left((BC)^2=\right)(p-12)^2+(q-2)^2$ or $(AC=)\sqrt{(p-12)^2+(q-2)^2}$ oe	M1
	$q=3p+6$	B1ft
	$\left((AB)^2=(AC)^2+(BC)^2\right)$	
	$\left("6\sqrt{10}"\right)^2=(p--6)^2+("3p+6"-8)^2+(p-12)^2+("3p+6"-2)^2$	ddM1A1
	$10p^2=80$ oe	M1
	$p=\sqrt{8}$ oe	A1
	$p=\sqrt{8}, q=6+3\sqrt{8}$ oe	[7]
(e)	Length $CX=\sqrt{\left("\sqrt{8}"-"0"\right)^2+\left("6+6\sqrt{2}"-"6"\right)^2}\left(=4\sqrt{5}\right)$	M1
	Area of triangle $ABC=\frac{1}{2}\times"4\sqrt{5}"\times"6\sqrt{10}"=60\sqrt{2}$ oe	dM1 A1 [3]
ALT	$\frac{1}{2}\begin{vmatrix}-6 & 12 & "\sqrt{8}" & -6\\ 8 & 2 & "6+3\sqrt{8}" & 8\end{vmatrix}$ oe	M1
	$\frac{1}{2}\left[\left(-6\times2+12\times"(6+3\sqrt{8})"+"\sqrt{8}"\times8\right)-\left(-6\times"(6+3\sqrt{8})"+"\sqrt{8}"\times2+12\times8\right)\right]$ oe	dM1
	$60\sqrt{2}$ oe	A1 [3]
		Total 17 marks

Type MS · Finding a point from a line and a distance

Answer 10 of 10 · Question 8 part d.ii · 7 marks

2023 JUN · Paper 02 · extract 3 of 5 · find part d.ii in this mark scheme extract

Part	Mark	Notes
(a)	M1	For correctly finding the gradient of AB in unsimplified form.
	dM1 (A1 on ePen)	For a full and correct attempt to find the equation of the line using their gradient. No simplification is required. If using $y = mx + c$, a value for c must be found.
	A1	For a correct equation in the required form.
(b)	M1	For using a correct method to find the length of line segment AB, in unsimplified form.
	A1	For the correct exact length.
(c)	B1 (M1 on ePen)	For either coordinate correct.
	B1 (A1 on ePen)	For both correct coordinates (0, 6) For part c the values of m and n can be explicitly identified or written in a coordinate.
In part (d) allow p to be interchangeable with x, q to be interchangeable with y throughout		
(d)	M1	For a correct statement of the gradient for either CA or CB
	M1	For using the negative perpendicular of either gradient and equating the gradients to form an equation in terms of p and q only.
	B1ft	For finding the negative reciprocal of their gradient of XC and placing this equal to a correct expression in p and q as shown.
	ddM1	For correctly substituting their **linear** expression for p or q into a **quadratic** equation in q or p to obtain an equation in one variable. Must use their $q = 3p + 6$ and is dependent on both previous method marks.
	A1	For the correct two term quadratic..
	M1	For **correctly solving their** quadratic to find a value for either p or q
	A1	For **both** p and q correct.
ALT1 (d)	M1	For the correct method to find the midpoint of AB
	M1	For the correct method to find the radius of C, ft their answer from part b if used.
	B1ft	For the equation of the line, unsimplified, ft their gradient of AB
	ddM1	For correctly substituting their $y = 3x + 6$ into the equation of a circle, using their midpoint of AB and their radius of C Must use their $y = 3x + 6$ and is dependent on both previous method marks.
	A1	Correct equation
	M1	For **correctly solving their** quadratic to find a value for either x or y
	A1	For **both** p and q correct.
ALT2 (d)	M1	For the correct method to find the length of AC or $(AC)^2$
	M1	For the correct method to find the length of BC or $(BC)^2$
	B1ft	For the equation of the line, unsimplified, ft their gradient of AB
	ddM1	For correctly substituting their lengths and their $q = 3p + 6$ into a correct Pythagorean equation, dependent on both previous method marks.
	A1	Correct equation
	M1	For **correctly solving their** quadratic to find a value for either p or q
	A1	For **both** p and q correct.

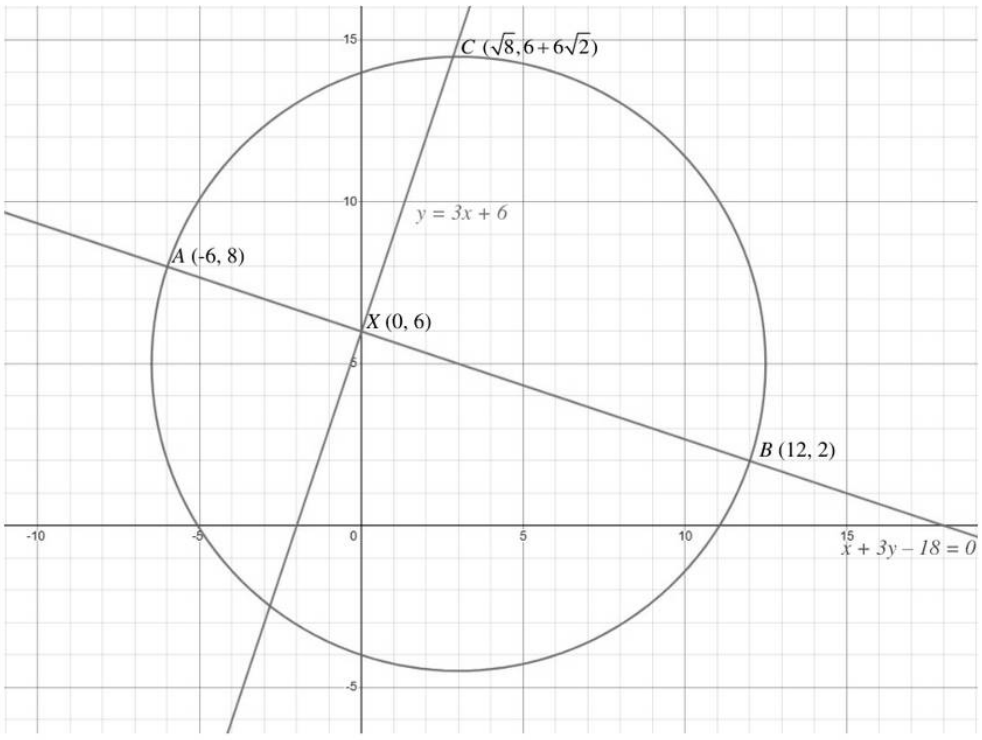
Type MS · Finding a point from a line and a distance

Answer 10 of 10 · Question 8 part d.ii · 7 marks

2023 JUN · Paper 02 · extract 4 of 5 · find part d.ii in this mark scheme extract

(e)	**M1**	For using a correct method to find the length of the perpendicular from AB to C
	dM1	For using their results from part (b) and their length of perpendicular from AB to C with the correct formula for the area of a triangle. Dependent on previous method mark.
	A1	For the correct area of $60\sqrt{2}$ or $30\sqrt{8}$ (units^2)
ALT	**M1**	For a correct statement for the area such as the one shown, using their values of p and q
	dM1	For the correct evaluation of their determinant Dependent on previous method mark.
	A1	For the correct area of $60\sqrt{2}$ or $30\sqrt{8}$ (units^2)

Useful Sketch for Parts c/d – look for any working on or near a sketch.



Type MS · Finding a point from a line and a distance

Answer 10 of 10 · Question 8 part d.ii · 7 marks

2023 JUN · Paper 02 · extract 5 of 5 · find part d.ii in this mark scheme extract

Question	Scheme	Marks